

Evaluating Evidence Citations in AI Generated Health Explanations

Preliminary research proposal for discussion

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Background and Motivation

I am interested in how generative AI can help people understand and check health information. My course-based MS in Information Systems and technical projects in retrieval-augmented generation (RAG), source citations, and model comparison provide a starting point for supervised research.

Problem and Research Question

An explanation can appear convincing even when its statements lack supporting evidence. Citations offer a way to check an answer, but a cited source may not support the specific claim. I propose to investigate: Do evidence citations help people recognise unsupported statements in AI-generated health explanations?

Scope

The initial study would focus on one narrow public health-education topic, such as basic diabetes terminology, using publicly available authoritative educational material. The prototype would provide general explanations rather than personalised diagnosis or treatment advice. The topic and question would be refined through literature review and supervision.

Proposed Method

First, review literature on healthcare LLM evaluation, citations, patient understanding, and human-AI interaction to refine the question and assess its contribution. Next, build a small RAG prototype and a documented question set. With healthcare expertise, evaluate answer correctness, citation support, and unsupported statements. If feasible, and after appropriate ethics review, compare matched explanations with and without citations in a participant study. The supervisor would guide sampling, study design, and analysis.

Evaluation and Expected Outputs

Technical evaluation would examine whether sources support the generated claims and where the system fails. A possible human study would measure comprehension, recognition of unsupported statements, and decisions to accept or verify information; increased trust alone would not count as success. Outputs would include a literature review, evaluation protocol, reproducible prototype, and research report. Publication would be considered only if the findings justify submission.

Supervision Sought

I am seeking guidance in healthcare accuracy, research design, human-centred evaluation, statistical analysis, and research ethics. I am open to refining this proposal or contributing to an existing project.

Current status: This is an initial proposal. The literature review, experiments, and participant study have not yet been conducted.